

What is MongoDB?

MongoDB is a NoSQL database..

NoSQL means:

- ★ Data is not stored in tables (rows and columns) like traditional databases.

Instead, MongoDB stores data in the form of documents.

What are Documents?

Documents are stored in a JSON-like format.

JSON is a simple format where data is stored in key-value pairs.

Example:

```
</> JSON  
  
{  
  "name": "Ali",  
  "age": 20  
}
```

- ★ "name" is the key and "Ali" is the value.
- ★ "age" is the key and "20" is the value.

Why is MongoDB Flexible?

In traditional SQL databases (like MySQL):

- ★ You must define a fixed structure (table with columns) before storing data

But in MongoDB:

- ★ There is no fixed structure required.

★ You can store different types of data.

```
</> JSON  
  
{ "name": "Ali", "age": 20 }  
  
{ "name": "Sara", "email": "sara@gmail.com" }
```

Both documents have different structures, but they can still be stored in MongoDB.

Real life example: Social Media App:

- ★ User 1: name + age.
- ★ User 2: name + email + phone.

In MongoDB, both users can be stored easily even though their data is different. In SQL databases, this can be a problem because the columns are fixed and must be the same for every user.

What does Scalable mean?

Scalable means a system can handle increasing data and users without slowing down or crashing.

For example, apps like Instagram and Amazon are scalable because they manage millions of users and large amounts of data smoothly.

Introduction:

MongoDB is a NoSQL database that stores data in flexible, JSON-like documents instead of tables, and it is designed to handle large, unstructured, and scalable data efficiently.

KEY FEATURES:

What are rich Queries?

Rich queries mean:

You can easily search data from the database using different conditions.

It is not just simple searching, but also includes:

- ★ Filtering
- ★ Sorting
- ★ Range searching

Example: If you have student data.

```
</> JSON  
  
{ "name": "Ali", "age": 20, "marks": 80 }  
{ "name": "Sara", "age": 22, "marks": 90 }
```

- ★ You can perform queries like:
- ★ Find students with marks > 85
- ★ Find students with age = 20

JSON-like Documents (MongoDB Feature)

What does it mean?

MongoDB stores data in JSON-like documents instead of tables, where each document contains data in key-value pairs.

Why is it useful?

- ✓ Easy to understand.
- ✓ Easy to modify.
- ✓ Flexible structure (each document can be different).

What is Indexing:

Indexing is used to organize data in a way that makes searching faster, just like an index in a book helps you find information quickly.

For example:

```
</> JSON  
  
{ "name": "Ali", "age": 20 }  
{ "name": "Sara", "age": 22 }  
{ "name": "Ahmed", "age": 25 }
```

Indexing = fast searching

Without indexing = slow searching

Which is better: MySQL (SQL) or MongoDB?

MongoDB is better when:

- ✓ The data structure changes frequently.
- ✓ Different users have different types of data.
- ✓ You are working with large amounts of data (Big Data)
- ✓ Real-time applications are needed.

Why MongoDB?

- It is flexible
- It is scalable
- It is fast for modern applications

MySQL is better when:

- ✓ The data structure is fixed
- ✓ The data is highly organized and structured
- ✓ Relationships between tables are important

Why MySQL?

- It has a strong fixed structure
- It ensures data accuracy
- It manages relationships easily

Both MongoDB and MySQL are powerful databases, but they are used for different purposes. MongoDB is better for flexible and large-scale applications like social media, while MySQL is better for structured and reliable systems like banking and management systems.

MongoDB vs SQL

MongoDB (Flexible Database)

User 1:

```
{ "name": "Ali",  
  age: 20  
}
```

User 2:

```
{ "name": "Sara",  
  email: "sara@gmail.com",  
  phone: "0300"  
}
```

- ✓ Different Structure Allowed
- ✓ No Fixed Columns
- ✓ Easy & Flexible

SQL (Traditional Database)

TABLE: Users

name	age	email	phone
Ali	20	NULL	NULL
Sara	NULL	sara@gmail.com	0300

- ✗ Fixed Structure
- ✗ Empty Values (NULL)
- ✗ Less Flexible